

C++ Beginners Training Manual

By GPT-5

1. Overview of C++

C++ is a high-level, general-purpose programming language created by Bjarne Stroustrup in the early 1980s.

It extends the C language with additional features such as object-oriented programming (OOP).

It's fast, powerful, and used in fields like gaming, finance, robotics, and system software.

2. Setting Up Your C++ Environment

1. Install a C++ Compiler (e.g., GCC or MinGW)
2. Use an IDE such as Code::Blocks, Visual Studio Code, or Dev-C++
3. Write and run your first program:

```
#include  
using namespace std;  
  
int main() {  
    cout << "Hello, World!" << endl;  
    return 0;  
}
```

3. Basic Syntax

- Variables and Data Types: int, float, char, string, bool
- Input/Output using cin and cout
- Operators: Arithmetic (+, -, *), Comparison (==, !=), Logical (&&, ||, !)

4. Control Structures

Use if/else, switch, for, while, and do-while to control program flow.

Example:

```
if (age >= 18)  
    cout << "Adult";  
else  
    cout << "Minor";
```

5. Functions

Functions make code modular and reusable.

Example:

```
int add(int a, int b) { return a + b; }
```

```
int main() { cout << add(3, 4); }
```

6. Arrays and Strings

Arrays store multiple values of the same type.

Example: `int scores[5] = {90, 85, 80, 75, 70};`

Strings store text: `string message = "Welcome to C++!"`;

7. Introduction to Object-Oriented Programming

Classes group data and behavior together.

Example:

```
class Student {  
public:  
    string name;  
    int age;  
    void introduce() {  
        cout << "Hi, I'm " << name << " and I'm " << age << " years old.";  
    }  
};
```

8. Practical Exercises

- Calculate area of a rectangle
- Display a multiplication table
- Build a simple calculator
- Store and display student grades

9. Tips for Beginners

- Practice small programs daily
- Understand logic before syntax
- Use comments to document your code
- Debug step by step — errors are part of learning!

10. Next Steps

Learn advanced topics like file handling, pointers, inheritance, and polymorphism.

Try building small projects like an inventory system or billing system.